47 An integrated map of microrelief and deposit thickness: the case of Cervia Vecchia (RA)

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Archaeological predictive maps are useful tools for both the research agenda and urban planning. In Italy, several projects have already addressed the systematisation of known archaeological evidence and the estimation of archaeological potential through open, and often open-source, WebGIS platforms and geo-databases, such as MAPPA (Pisa), RAPTOR, SITAR (Rome) and ArcheoDB (Emilia-Romagna). However, none of these projects publishes a fully open-source (FLOSS) and reproducible workflow capable of extending point-based stratigraphic evidence (deposit thickness and depth) across an entire study area, nor do they formally test the covariates chosen to support such an interpolation. This paper aims to address this gap by proposing and testing a reproducible, entirely FLOSS-based methodology for developing the auxiliary cartographic elaborations needed for archaeological potential maps, conceived to be applicable beyond a single case study. The methodology is tested through the case study of Cervia Vecchia, an abandoned medieval town that has been the object of an excavation promoted by the University of Bologna since 2019. Specifically, different categories of datasets are cross-referenced in order to develop: a microrelief map, aimed at individuating topographic anomalies of archaeological value, and an archaeological depth map, covering the entire study area thanks to geostatistical interpolation (Ordinary Kriging). The work focused on methodological and computational aspects, like raster analysis and sampling, defining parameters for variograms, aimed at developing two complementary products: a micro relief map and a thickness and depth map. Additionally, this work tested the implementation of covariates, based on morphological evidence and ceramic findings, in order to extend the predictive model (Regression Kriging and Co-Kriging). Also, based on the statistical independence between the two datasets, this work tried to implement a preliminary evaluation of the "archaeological potential", built thanks to raster calculator functions based on a spatial co-occurrence criterion. Relying exclusively on open-source tools throughout, the proposed workflow is intended to be transferable to other archaeological contexts, offering fieldworkers and heritage authorities a transparent, low-cost and reproducible methodology to fill the gap outlined above and to support the development of future archaeological potential maps.